VASP Analyzer

Installation Guide (macOS)

VASP Analyzer is a desktop application for visualising and editing VASP calculations — crystal structures, band structures, densities of states, charge densities, molecular-dynamics trajectories, NEB reaction paths and phonons.

Nothing else to install. The download is a self-contained bundle: Python, Qt and every other dependency are already inside it. You do not need Python, pip, conda or VASP itself installed to run the app.

Before you start

- **macOS 11 (Big Sur) or newer.**
- **An Apple Silicon Mac** — any M1, M2, M3 or M4 model.
- About **600 MB** of free disk space once unzipped.

Intel Macs are not supported by this build. This package contains an Apple Silicon (arm64) binary only, and will not launch on an Intel Mac. To check which you have, click the Apple menu → *About This Mac*: if the *Chip* line says *Apple M1/M2/M3/M4* you are fine; if it says *Intel*, ask for an Intel build instead.

Installing

1. **Unzip** `VASP Analyzer-macOS.zip`. This produces `VASP Analyzer.app`.
2. **Drag** `VASP Analyzer.app` into your **Applications** folder. Optional, but it keeps the app out of Downloads and makes it appear in Spotlight and Launchpad.
3. **First launch only: right-click** (or Control-click) the app and choose **Open**, then confirm **Open** in the dialog. After this first time, double-clicking works normally.

Why the right-click is necessary. The app is not signed with a paid Apple Developer certificate, so macOS blocks it on first launch as coming from an "unidentified developer". Right-click → Open is the standard way to approve it. Simply double-clicking the first time will show a refusal with no Open button.

If macOS says the app is "damaged and can't be opened"

That message is misleading — the app is fine. macOS attaches a quarantine flag to anything downloaded from the internet, and for unsigned apps it sometimes reports this instead of the normal warning. Clear the flag by opening **Terminal** and running:

```
xattr -cr "/Applications/VASP Analyzer.app"
```

If you put the app somewhere other than Applications, use that path instead. Then open the app again as normal.

Opening your first calculation

There are three ways in:

- **Drag and drop** a calculation folder onto the window.
- **File → Open Calculation…** (Ctrl+O) to pick a folder.
- **File → Open File…** (Ctrl+Shift+O) for a single file.

A folder holding VASP files directly works as-is. So does a project folder organised into `band`, `md`, `neb` and `phonon` subdirectories — each page finds its own data independently, and a project with only some of them is fine.

Single files work too: a lone **POSCAR** opens the Structure and Editor pages, and a lone **CHGCAR** opens those plus Charge Density.

Nothing is modified. The app reads your calculation files and never writes to them. The only files it creates are ones you explicitly save or export yourself.

Where to go next

See **user_guide.pdf** for what each page does, how to read the plots, and the full list of keyboard shortcuts.

If something does not work

- A page saying data is unavailable is normal — it names the file it needs (EIGENVAL for bands, DOSCAR for the density of states, and so on). The app configures each page from whatever your calculation actually contains.
- For anything else, **Help → Show Log File** opens the log, which records what was parsed and what failed. Sending that log is the fastest way to get a problem diagnosed.